

Malware Detection Using Artificial Intelligence: A Survey



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Abstract

Malware has emerged as one of the most critical cybersecurity threats in the digital era, causing significant financial and data losses worldwide. Traditional malware detection techniques, such as signature-based and heuristic-based methods, are increasingly ineffective against modern threats like polymorphic and zero-day malware. Artificial Intelligence (AI), particularly Machine Learning (ML) and Deep Learning (DL), offers a promising solution by enabling systems to automatically learn patterns and detect unknown threats. This paper provides a comprehensive survey of AI-based malware detection techniques. It explores various ML and DL approaches, including supervised, unsupervised, and hybrid models. Additionally, the paper discusses feature engineering techniques, datasets, evaluation metrics, and real-world applications. The challenges of AI-based detection, such as adversarial attacks and data imbalance, are also analyzed. Finally, future research directions, including explainable AI and federated learning are presented.

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Introduction

Malware is one of the most serious and rapidly evolving threats in cyber-security, affecting individuals, organizations, and governments worldwide. It includes various types such as viruses, ransom-ware, spyware, and Trojan, all designed to disrupt systems, steal sensitive data, or cause financial damage. As digital systems become more interconnected, the impact of malware attacks has grown significantly, often leading to large-scale data breaches, operational shutdowns, and economic loss.

Traditional detection methods, which rely mainly on known signatures, are no longer sufficient to handle modern threats, especially new and unknown (zero-day) attacks. Artificial Intelligence offers a more advanced solution by using machine learning and behavior-based analysis to detect unusual patterns and anomalies in real time. This allows systems to identify and respond to previously unseen malware, making AI a powerful and adaptive tool in modern cyber security defense.

Background

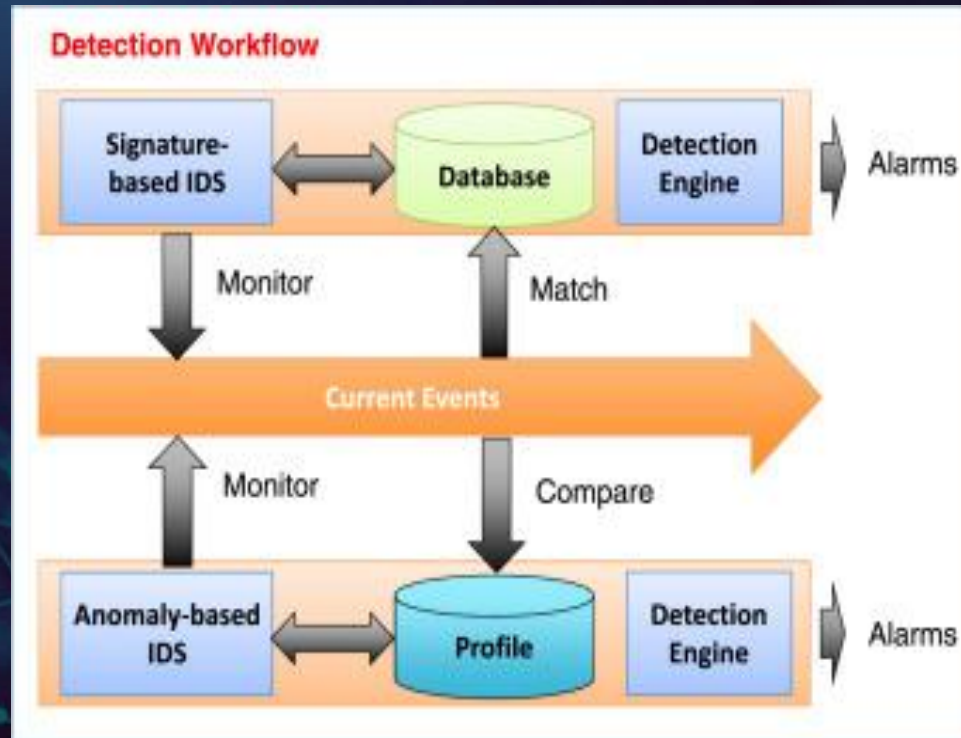


Figure 1: Signature-based Detection

Signature-based detection matches files or traffic against a database of known threat signatures (hashes, strings).

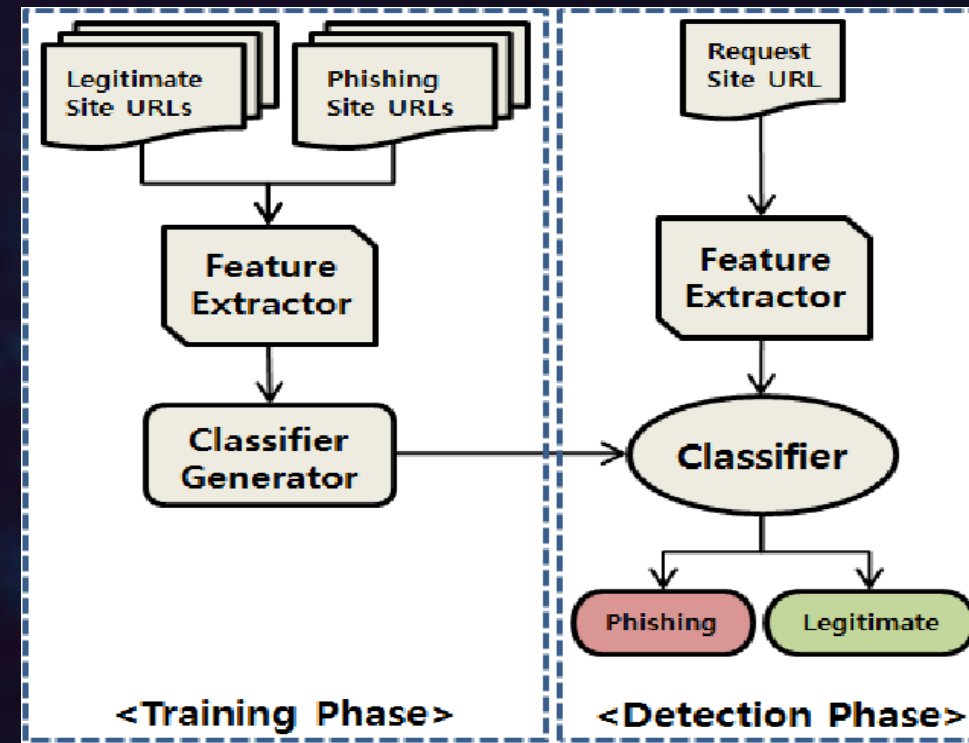


Figure 2: Heuristic Methods

Heuristic method employs rules or algorithms to detect suspect characteristics, such as code analysis that resembles known malicious families without an exact signature match.

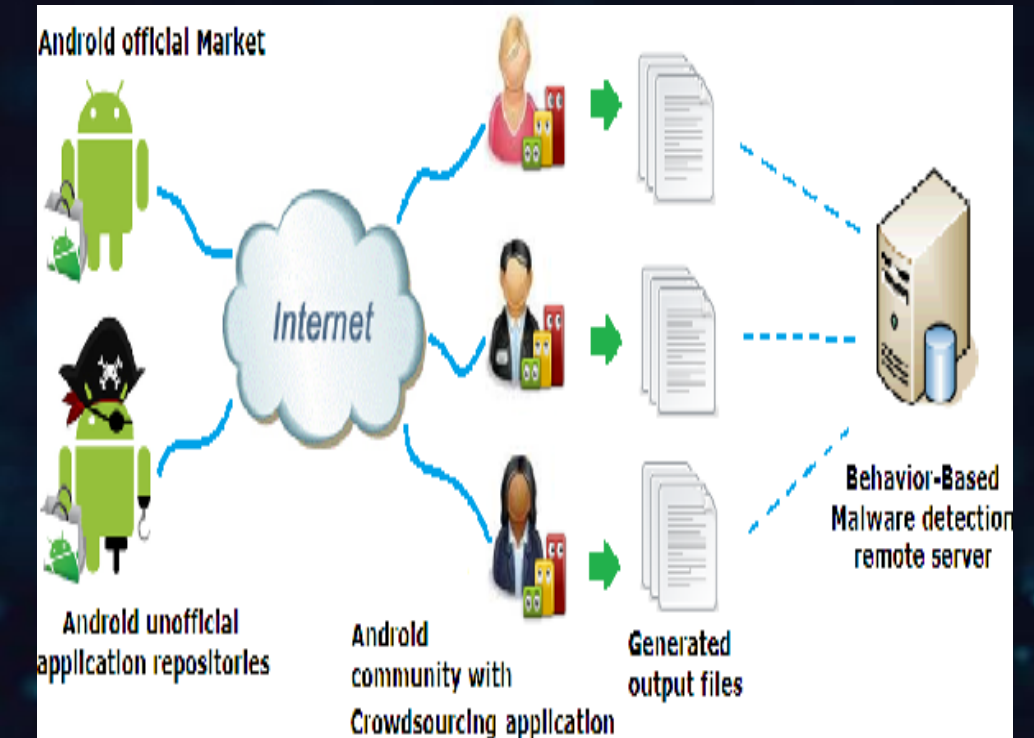


Figure 3: Behavior-based Systems

For monitor-based systems, monitors system activities in real-time to identify anomalous behavior, rather than focusing on the file's static code.

Types of Malware



Virus

Attaches itself to legitimate files and spreads when executed



Worm

Self-replicates and spreads across networks without user interaction



Spyware

Collects sensitive user information without consent

Trojan Horse

Disguised as legitimate software but performs malicious actions



Ransom-ware

Encrypts user data and demands payment for the decryption



Adware

Displays unwanted advertisements



Rootkits

Provide unauthorized access while hiding their presence



Literature Review

Journal of Machine Learning Research 7 (2006) 2721-2744

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Learning to Detect and Classify Malicious Executables in the Wild*

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Kolter and Maloof used machine learning for detecting malicious executable, showing improved detection accuracy.

Deep Neural Network Based Malware Detection Using Two Dimensional Binary Program Features

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Abstract

Malware remains a serious problem for corporations, government agencies, and individuals, as attackers continue to use it as a tool to effect frequent and costly network intrusions. Today malware detection is still done mainly with heuristic and signature-based methods that struggle to keep up with malware evolution. Machine learning holds the promise of automating the work required to detect newly discovered malware families, and could potentially learn generalizations about malware and benign software (benignware) that

machine learning classification model, with false positive rates that approach traditional labor intensive signature based methods, while also detecting previously unseen malware. Since machine learning models tend to improve with larger data-sizes, we foresee deep neural network classification models gaining in importance as part of a layered network defense strategy in coming years.

1. Introduction

Saxe and Berlin applied deep neural networks for malware classification using binary features.

Malware Detection by Eating a Whole EXE

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Abstract

In this work we introduce malware detection from raw byte sequences as a fruitful research area to the larger machine learning community. Building a neural network for such a problem presents a number of interesting challenges that have not occurred in tasks such as image processing or NLP. In particular, we note that detection from raw bytes presents a sequence problem with over two million time steps and a problem where batch normalization appear to hinder the learning process. We present our initial work in building a solution to tackle this problem, which has linear complexity dependence on the sequence length, and allows for interpretable sub-regions of the binary to be identified. In doing so we will discuss the many challenges in building a neural network to process data at this

inside a specially instrumented environment, such as a customized Virtual Machine (VM), which introduces high computational requirements. Furthermore, in some cases it is possible for malware to detect when it is being analyzed. When the malware detects an attempt to analyze it, the malware can alter its behavior, allowing it to avoid discovery (Raffetseder, Kruegel, and Kirda 2007; Garfinkel et al. 2007; Carpenter, Liston, and Skoudis 2007). Even when malware does not exhibit this behavior, the analysis environment may not reflect the target environment of the malware, creating a discrepancy between the training data collected and real life environments (Rossow et al. 2012). While a dynamic analysis component is likely to be an important component for a

Raff et al. proposed feeding raw executable files into deep learning models, eliminating manual feature engineering.

ML improves detections

DL models outperform ML models

DREBIN: Effective and Explainable Detection of Android Malware in Your Pocket

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Abstract

Malicious applications pose a threat to the security of the Android platform. The growing amount and diversity of these applications render conventional defenses largely ineffective and thus Android smartphones often remain unprotected from novel malware. In this paper, we propose DREBIN, a lightweight method for detection of Android malware that enables identifying malicious applications directly on the smartphone. As the limited resources impede monitoring applications at run-time, DREBIN performs a

tionality to its users. Unfortunately, smartphones running Android are increasingly targeted by attackers and infected with malicious software. In contrast to other platforms, Android allows for installing applications from unverified sources, such as third-party markets, which makes bundling and distributing applications with malware easy for attackers. According to a recent study over 55,000 malicious applications and 119 new malware families have been discovered in 2012 alone [18]. It is evident that there is a need for stopping the proliferation of malware on Android markets and smartphones.

DL4MD: A Deep Learning Framework for Intelligent Malware Detection

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Abstract- *In the Internet-age, malware poses a serious and evolving threat to security, making the detection of malware of utmost concern. Many research efforts have been conducted on intelligent malware detection by applying data mining and machine learning techniques. Though great results have been obtained with these methods, most of them are built on shallow learning architectures, which are still somewhat unsatisfying for malware detection problems. In this paper, based on the Windows Application Programming Interface (API) calls extracted from the Portable Executable (PE) files, we study how a deep learning architecture using the stacked AutoEncoders (SAEs) model can be designed for intelligent malware*

attackers through the techniques such as encryption, polymorphism and obfuscation [29], [2]. Furthermore, malicious files are being disseminated at a rate of thousands per day [34], making it difficult for this signature-based method to be effective. In order to combat the malware attacks, intelligent malware detection techniques need to be investigated. As a result, many researches have been conducted on intelligent malware detection by applying data mining and machine learning techniques in recent years [27], [1], [23], [32], [35].

Arp et al. introduced the DREBIN system for Android malware detection.

Hardy et al. proposed DL4MD, a deep learning framework for intelligent malware detection.

Hybrid(ML + DL) approaches are best.

Machine Learning Techniques

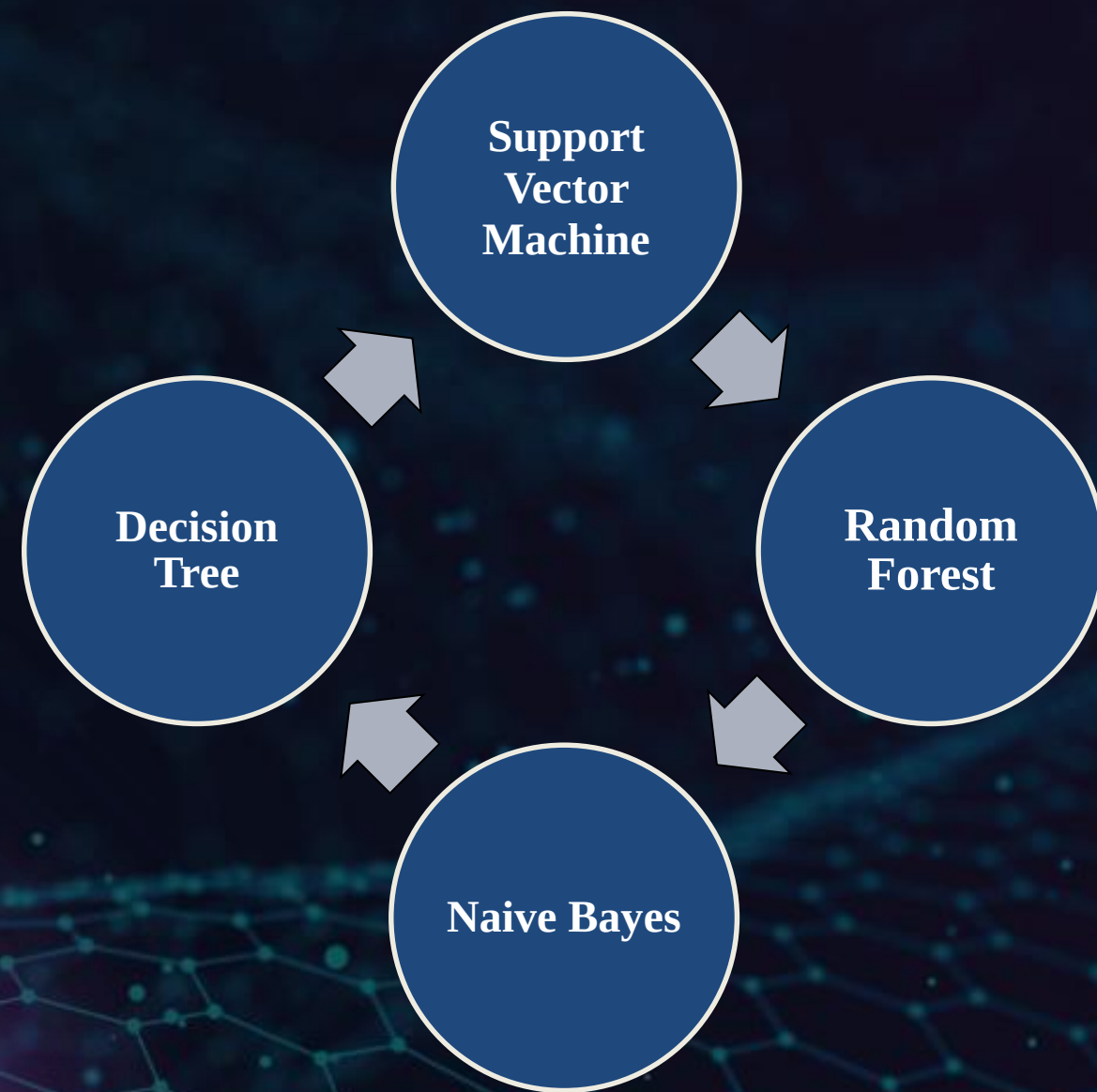


Figure 4: Supervised Learning Techniques

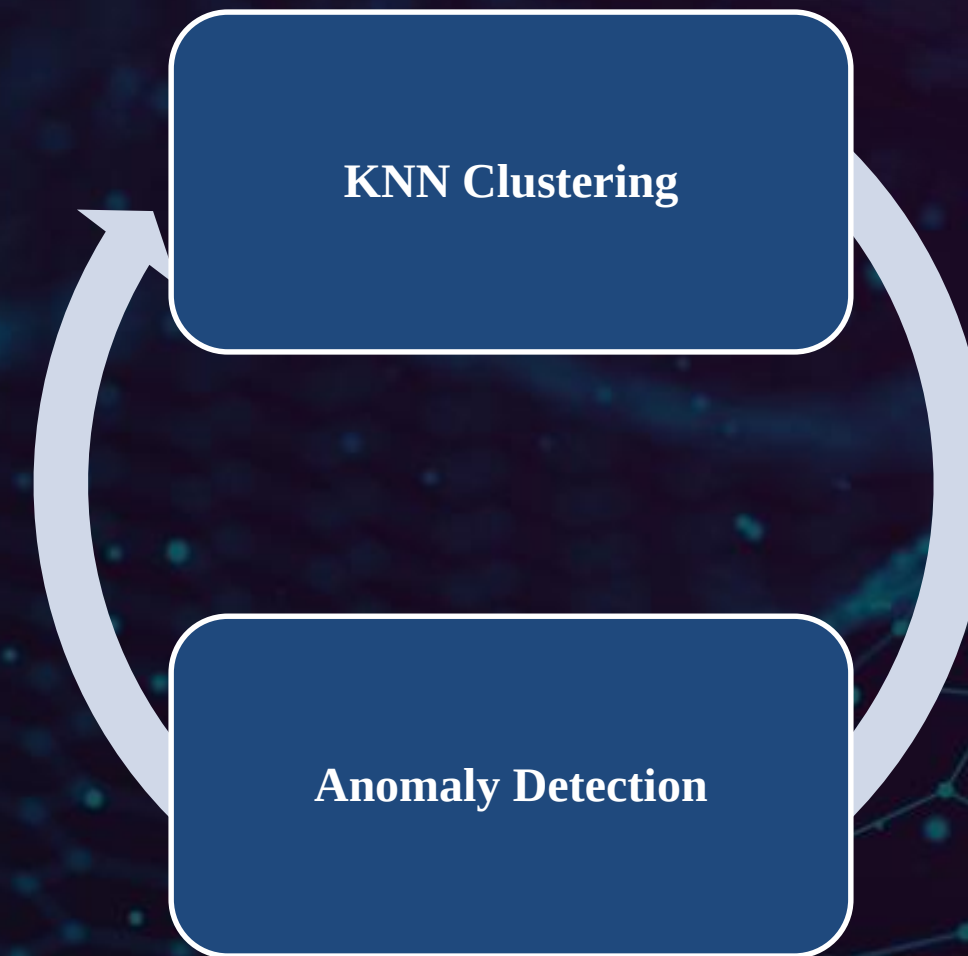
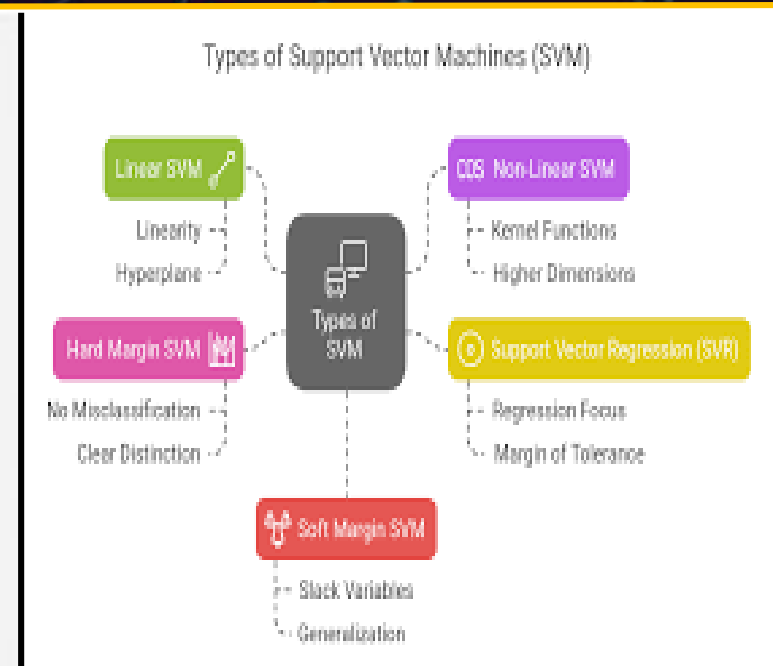
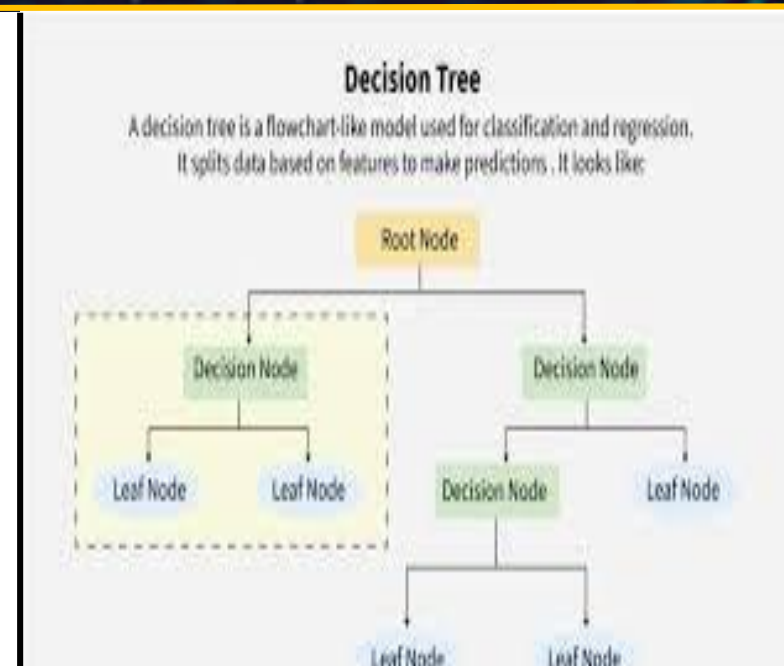
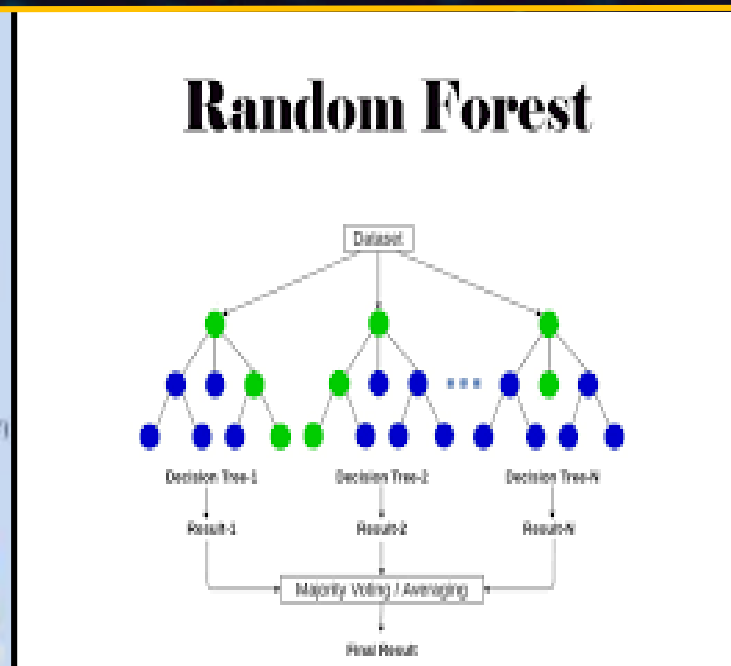
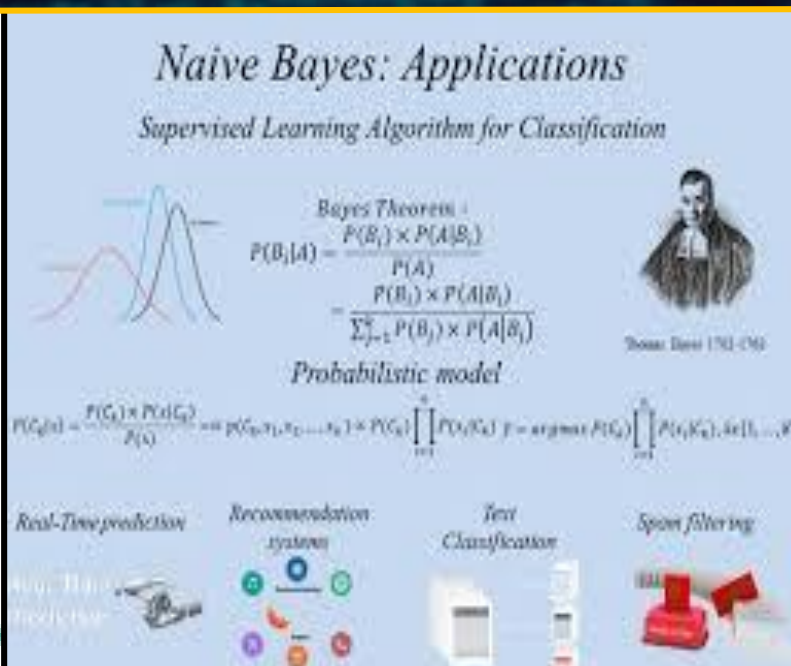
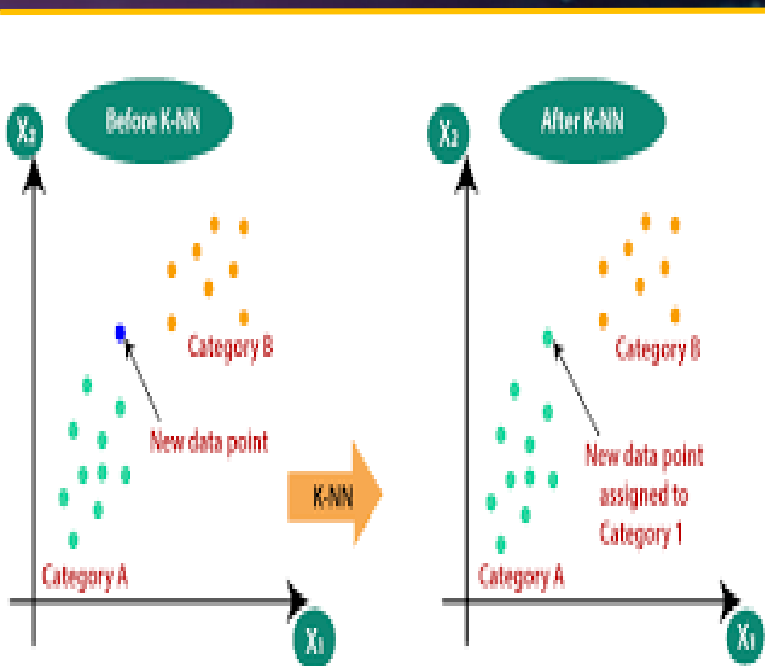


Figure 5: Unsupervised Learning Techniques

Algorithm	Type	Use of Performance	Case
SVM	Supervised	Binary classification	High
Decision Tree	Supervised	Rule-based classification	Medium
Random Forest	Ensemble	Large dataset classification	High
Naïve Bayes	Probabilistic	Fast classification	Medium
KNN Clustering	Instance-Based	Similarity detection	Low

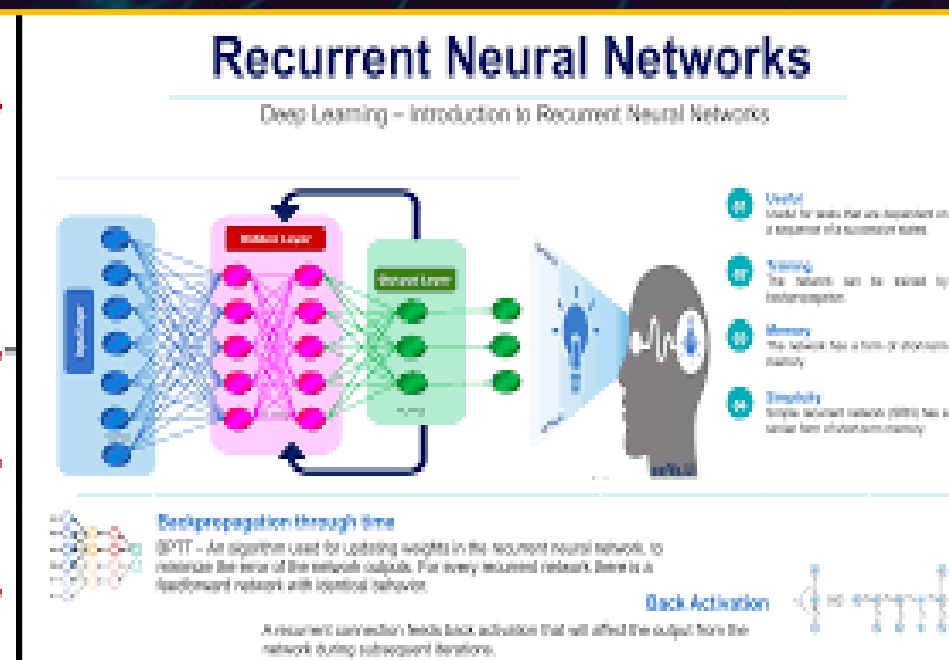
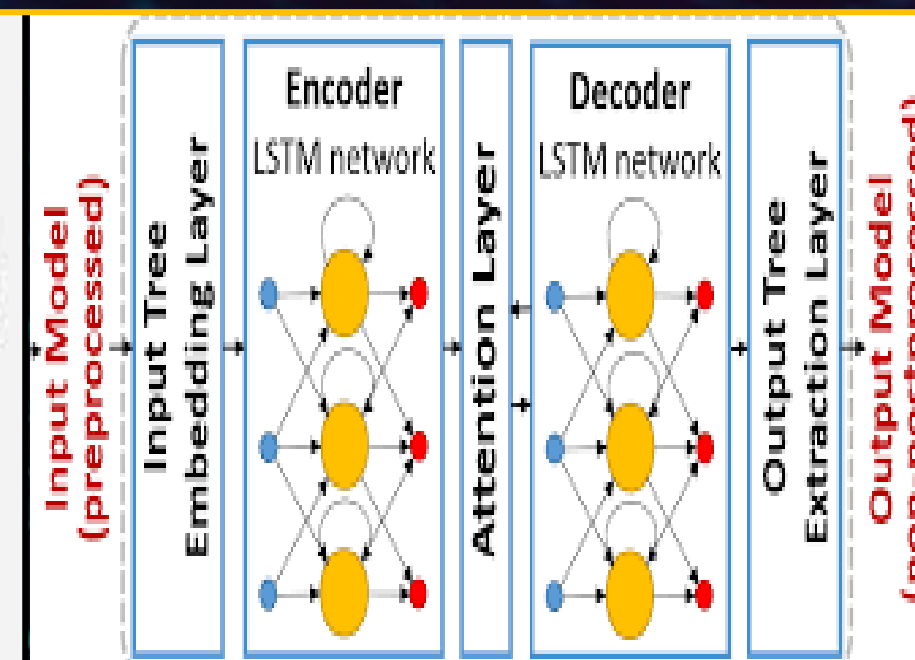
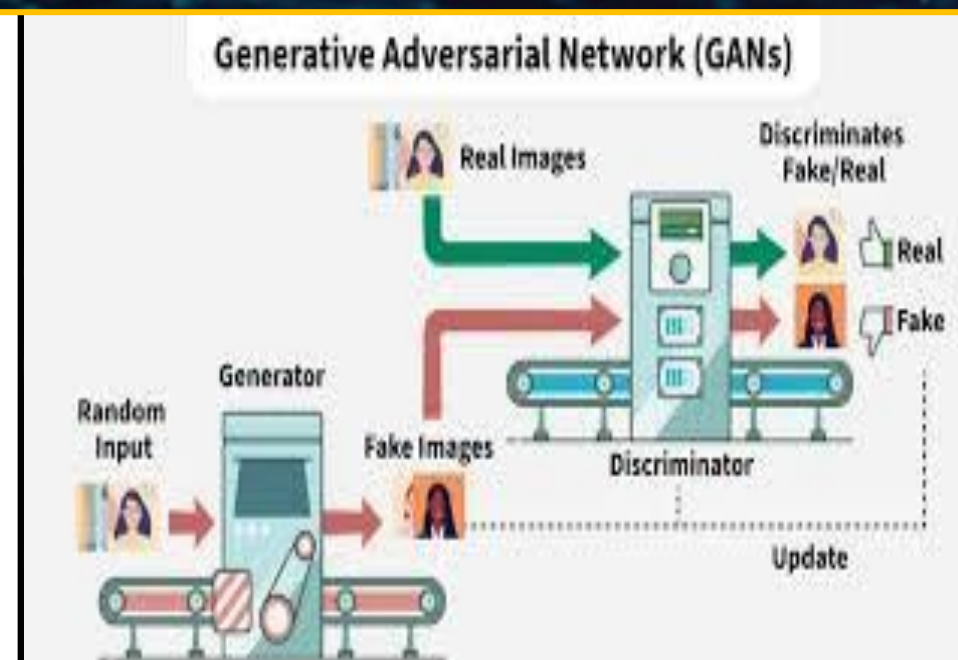
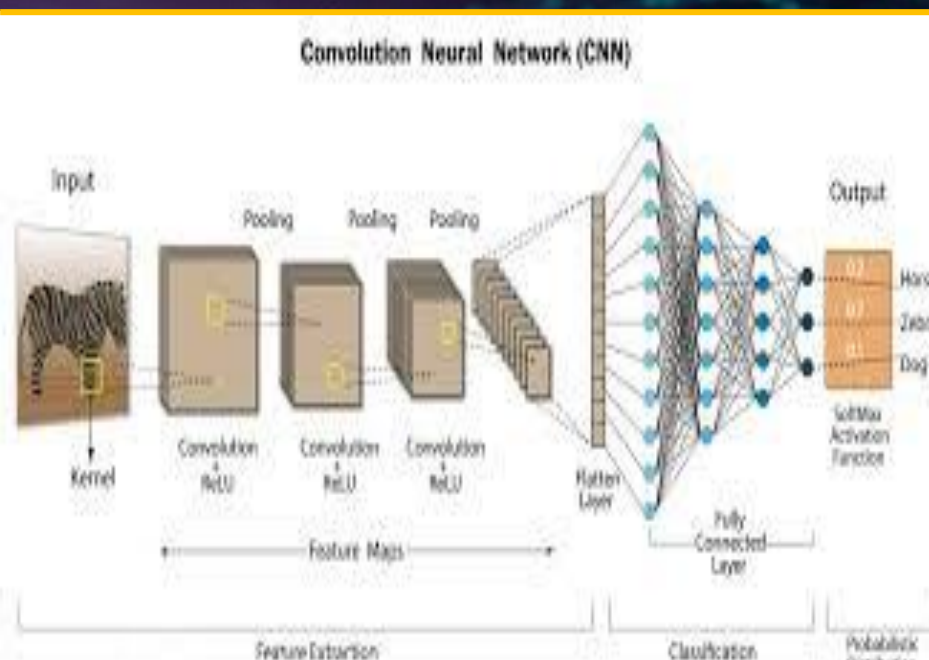
Table 1: Machine Learning Algorithms for Malware Detection



Deep Learning Techniques

	CNN = Convolutional Neural Networks	
	RNN = Recurrent Neural Network	
	LSTM = Long Short-Term Memory	
	GAN = Generative Adversarial Network	

Table 2: Deep Learning Algorithms for Malware Detection



The AI-based Hybrid Framework

A hybrid AI-based framework is proposed and performed by the researchers combining both static and dynamic analysis.

Key features

- ML + DI + AI
- Hybrid feature extraction
- Real time detection

Framework Steps

- Static and dynamic analysis
- Feature fusion
- Classification using ensemble model

Advantages

- Improved accuracy
- Reduced false positives
- Robust against obfuscation

Environment	Dataset Split	Models Used
Programming Language: Python	Training: 70%	Random Forest
Tools: TensorFlow, Scikit-learn	Testing: 30%	Support Vector Machine
Hardware: GPU-enabled system	Total: 100%	CNN and ANN

Table 3: Experimental Setup

System Architecture

Common datasets used in malware detection include:

- EMBER Dataset
- Virus-Share
- CICIDS
- Microsoft Malware Dataset

The key Challenges are:

- Imbalanced data
- Lack of standard datasets

CICIDS and Virus-share dataset are shown below:

1.4.4	3	0	4
1.4.4	3	0	4
1.4.4	3	0	4
1.4.4	3	0	4
1.4.4	3	0	4
1.4.4	3	0	4
1.4.4	3	0	4
1.4.4	3	0	4
1.4.4	3	0	4
1.4.4	3	0	4

0	0	14/02/2018 08:31:01	112641719
0	0	14/02/2018 08:33:50	112641466
0	0	14/02/2018 08:36:39	112638623
22	6	14/02/2018 08:40:13	6453966
22	6	14/02/2018 08:40:23	8804066
22	6	14/02/2018 08:40:31	6989341
0	0	14/02/2018 08:39:28	112640480



Figure 6: Workflow of the Methodology

Architecture Components/Methodology:

1. **Data Collection Module/Input:** Collects raw data such as executable files, network traffic, and system logs.

2. **Preprocessing Module:**

- Data cleaning
- Noise removal
- Normalization

3. **Feature Extraction Module:** Extracts relevant features such as:

- ❑ API call sequences
- ❑ Opcode frequencies
- ❑ File metadata

4. **Model Training Module:** Machine Learning and Deep Learning models are trained using labeled datasets.

5. **Detection Engine:** Classifies incoming files as:

- ❖ Benign
- ❖ Malicious

6. **Response Module:** Takes action such as:

- Blocking file
- Alerting user
- Logging attack

Results and Discussions

The experimental results demonstrate that AI-based models outperform traditional methods. From Observations we notice that Deep Learning models achieve higher accuracy than Machine Learning models. Also, Artificial Neural Network has an important role in increasing the accuracy. Hybrid models reduce false positives and feature quality significantly impacts performance. Performance is measured using Accuracy, Precision, Recall, F1-score, ROC-AUC and False Positive Rate.

Method	Accuracy	Advantages	Limitations
Signature-Based	Low	Fast	Cannot detect new malware
Machine Learning	Medium	Detects unknown threats	Feature dependent
Deep Learning	High	Automatic feature learning	High computation cost
Hybrid Approach	Very High	Best performance	Complex implementation

Table 3: Comparative Analysis of the Methods

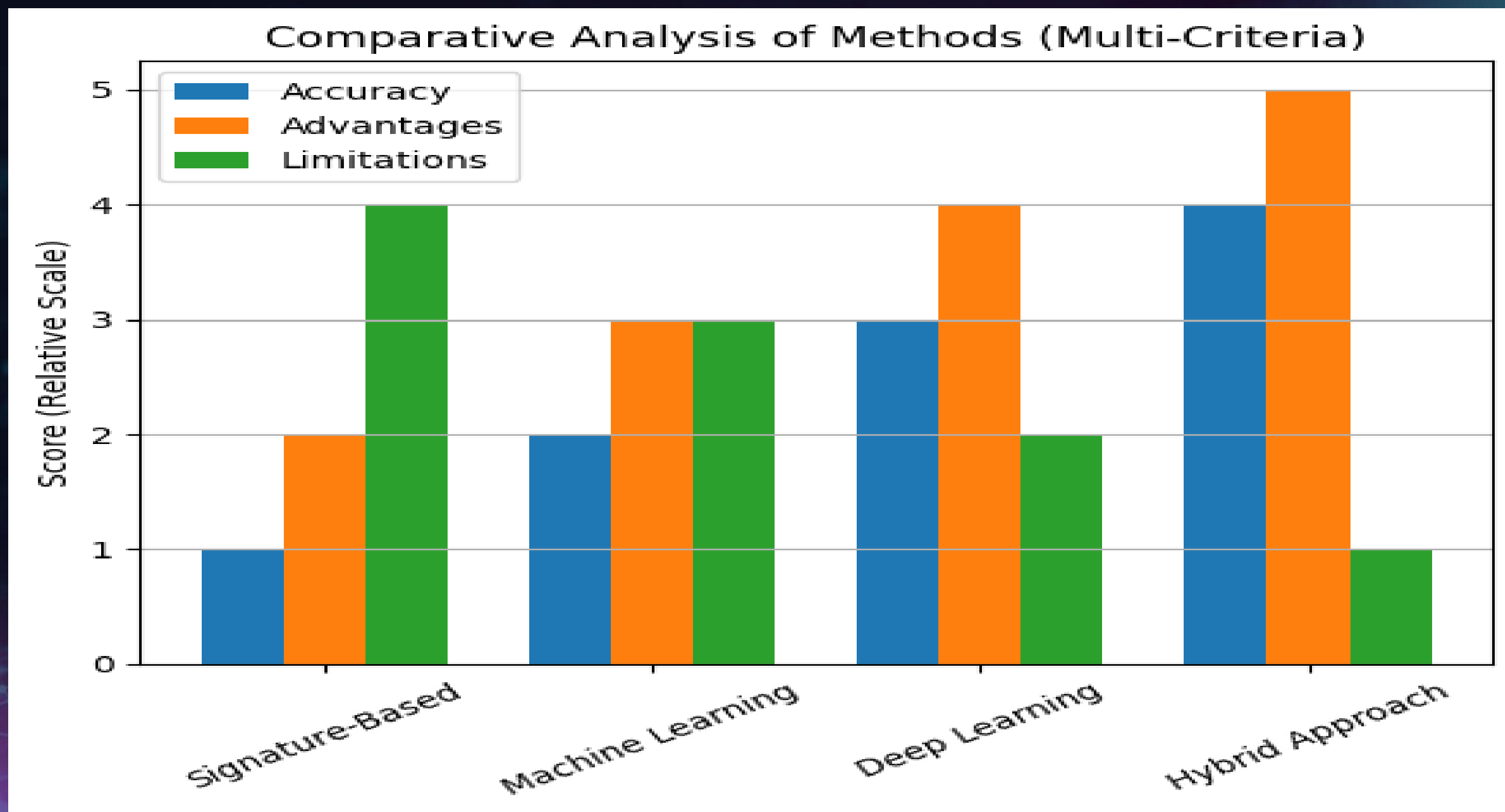


Figure 6: Comparative Analysis of Methods(Multi-criteria)

Case Study

Topic



Detecting ransom-ware
using AI-based models

Process



Dataset collected, feature
extracted, model trained
using supervised learning

Outcome



Accuracy improved, Data
encryption was prevented

Limitations & Challenges

Data imbalance

Model generalization issues

High computational cost

Lack of Explain-ability

Limited dataset availability

High computational requirements



Conclusion and Future Work

AI significantly improves malware detection by analyzing large volumes of data and identifying patterns that traditional methods may miss. It can detect new and unknown threats (zero-day attacks) using behavior-based analysis rather than relying only on known signatures. However, challenges still exist, such as false positives, vulnerability to adversarial attacks, high computational costs, and the need for large, high-quality datasets. Therefore, further research is required to make these systems more accurate, efficient, and secure for real-world applications.

Emerging approaches like Explainable AI (XAI), Federated Learning, and Edge AI are helping to address these issues. Explainable AI makes model decisions transparent and understandable, increasing trust in AI systems. Federated Learning enhances privacy by training models across multiple devices without sharing sensitive data, while Edge AI enables real-time malware detection directly on local devices, reducing latency and improving data security. Together, these technologies represent promising directions for the future of AI-based cyber security.

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THANK YOU!

